



PH 002: Preparatory Physics - 2

Course Code	PH 002
Course Name	Preparatory Physics - 2
L-T-P-C	3-1-0-4
Intended for	Preparatory Students
Distribution	Core
Approval	50th BoA (Jul 2023)

Course Contents

- **Electrostatics:** Electric Charge, Coulomb's law, electric field and electric potential, electric field lines, electric dipole, Gauss' law, capacitors and capacitance, current, resistivity, Ohm's law, resistance in series and parallel, Kirchhoff's law. [9 Lectures]
- **Magnetism:** Magnetism, magnetic field, Bio Savart's law, Ampere's law, Faraday's laws of induction, self and mutual inductance, LCR Circuit. [9 Lectures]
- **Optics:** Reflection of light, spherical mirrors, mirror formula. Refraction of light, total internal reflection and its applications, lenses, thin lens formula, magnification power of a lenses, wave nature of light, Huygen's principle, interference, diffraction, Young's double slit experiment. [9 Lectures]
- **Modern Physics:** Structure of the atom, Bohr's model, alpha, beta and gamma radiations, law of radioactive decay, half-life and mean life, blackbody radiation, Wien's law, Stefan's law, photoelectric effect, X-rays, Moseley's law, de Broglie wavelength of matter waves, ideas of quantum physics. [15 Lectures]

Text Books

1. D. Halliday, R. Resnick, and J. Walker, **Fundamentals of Physics**, John Wiley.
2. Sears and Zemansky's **University Physics with Modern Physics**, 13th Edition, Pearson.